WATER PURIFICATION

Finding water in an emergency and making it safe to drink: boiling, bleach, filters and tablets.

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GUIDE / VERIFY: General preparedness information, not a substitute for professional training, local rules, manufacturer instructions or medical care. **Practice skills in safe conditions before you need them, and verify important information with current official sources.**

IF OFFICIALS ISSUE A BOIL-WATER ADVISORY OR "DO NOT DRINK" NOTICE, FOLLOW IT. It tells you exactly what is wrong with the water and what works. Some contamination (fuel, chemicals) cannot be fixed by boiling or bleach: use bottled water instead.

1. WHERE TO FIND WATER

INSIDE YOUR HOME (USUALLY SAFE)

- Stored water. *(TL-02)*
- The water heater tank. Turn off its power or gas first, let it cool, open the drain at the bottom.
- Water in your pipes: open the highest faucet in the house to let air in, then drain from the lowest one.
- Ice cubes; liquid in canned fruit and vegetables.
- The toilet **tank** (not the bowl), only if no chemical cleaners are used in it. Treat it first.

OUTSIDE (ALWAYS TREAT)

- Rainwater, streams, rivers, ponds, lakes, springs.
- Moving water is usually better than still water.
- Collect upstream of roads, farms, buildings and animals.

NEVER DRINK

- Flood water.
- Water from radiators, boilers, water beds, pools or hot tubs.
- Water that smells of fuel or chemicals, even after treatment.

2. FIRST: CLEAR IT

Cloudy water protects germs from heat and chemicals. Before you treat it:

1. Let it stand until the sediment settles.
2. Pour the clear water off the top, or strain it through a clean cloth, paper towel or coffee filter.
3. Then treat it with one of the methods below.

3. FOUR WAYS TO MAKE IT SAFE

METHOD	HOW	KNOW THIS
Boiling *(best)*	Bring clear water to a **rolling boil for 1 minute**. Above 6,500 feet elevation, boil for **3 minutes**. Let it cool; cover it.	Kills germs. Does not remove chemicals. Pour it back and forth between two clean containers to improve the flat taste.
Household bleach	Use plain, **unscented** chlorine bleach (5% to 9% sodium hypochlorite). Add **8 drops per gallon** of clear water, or **16 drops** if it's cloudy or very cold. Stir; **wait 30 minutes**. It should smell slightly of chlorine; if not, repeat the dose and wait 15 more minutes.	Check the label: if it gives directions for disinfecting water, follow those. Never use scented, color-safe, splash-less bleach or bleach with added cleaners. Bleach loses strength over time; replace it yearly.

Purification tablets	Chlorine dioxide or iodine tablets. Follow the package directions exactly, including the waiting time.	Chlorine dioxide works against more germs. Pregnant women and people with thyroid problems should not use iodine.
Filters	Use a filter rated to remove protozoa and bacteria (pore size of 1 micron or smaller). Follow the maker's instructions.	Most portable filters **don't** remove viruses. Filter, then disinfect with bleach or tablets, for the best protection.

4. BLEACH QUICK CHART

For bleach that is 5% to 9% sodium hypochlorite. A standard dropper or eyedropper gives about 20 drops per milliliter.

AMOUNT OF WATER	CLEAR WATER	CLOUDY OR VERY COLD	WAIT
1 quart / 1 liter	2 drops	4 drops	30 minutes
1 gallon	8 drops	16 drops	30 minutes
5 gallons	40 drops (about 1/2 teaspoon)	80 drops (about 1 teaspoon)	30 minutes

5. KEEPING IT SAFE

- Store treated water in clean, covered containers. Label them "treated" with the date.
- Wash your hands before handling drinking water.
- Under a boil-water advisory, use safe water for drinking, cooking, brushing teeth, making ice, washing produce and mixing baby formula.
- Keep drinking water separate from water for flushing or washing.
- Pour water out; don't dip cups into the container.
- If someone gets diarrhea, vomiting or a fever after drinking untreated water, see a doctor.

6. PRACTICE RECORD

DATE	SOURCE	METHOD	NOTES (TASTE, TIME, PROBLEMS)

SOURCES / FURTHER REFERENCE

1. Centers for Disease Control and Prevention: How to Make Water Safe in an Emergency; Water Treatment Options When Hiking, Camping or Traveling. `cdc.gov/water-emergency`
2. U.S. Environmental Protection Agency: Emergency Disinfection of Drinking Water. `epa.gov/ground-water-and-drinking-water`
3. FEMA: Ready.gov: Water. `ready.gov/water`

IMPORTANT: These methods reduce germs; none of them removes fuel, chemicals or heavy metals. Follow your local water utility and health department. This document reflects the sources above as of November 2026.